

Automatic Identification of Jordanian Arabic Dialects Using Character N-Gram Features and LinearSVC

Pattern Recognition Project Report

Abstract

Arabic dialect classification is a challenging task within Natural Language Processing (NLP) because of logistic diversity, informal spelling versions, and also the absence of publicly available datasets for regional dialect identification. Existing Arabic NLP datasets usually categorize Jordan as one dialect label, overlooking local dialectal variation. This paper presents a machine learning framework focused on local Jordanian dialects. The classification is done using character-level TF-IDF features and multiple classifiers. The suggested system classifies text into four Jordanian dialect categories: Ammani, Bedouin, Karaki, and Fallahi.

The dataset was constructed by collecting from Jordanian social media sources, moreover preprocessing steps were taken on this dataset such as normalization, deduplication, and text cleaning. Three machine learning models were utilized for this project: Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest. The experimental outcomes demonstrate that SVM achieved the best overall performance with superior macro F1-score and classification accuracy. Moreover, confusion matrix and error analysis show major overlap between geographically similar dialects, highlighting the complexity of local linguistic variation.

Keywords— Arabic NLP, Pattern Recognition, Dialect Classification, Machine Learning, TF-IDF, Linear SVM

I. INTRODUCTION

Arabic is a macrolanguage that is spoken by over 400 million speakers in the Arab world. Modern Standard Arabic (MSA) is the formal written register, however the daily spoken and informal written communication of Arab populations is taken over by multiple regional dialects which differ essentially in multiple aspects including vocabulary, morphology, phonology, and syntax [1]. These dialectal adaptations pose a great challenge for natural language processing (NLP) tasks such as machine translation, sentiment analysis, and information retrieval, which normally assume a homogeneous language model.

Jordan has a variety of different regional dialects, spoken by different communities and in different areas of the country. To begin with, the Ammani dialect is spoken predominantly in the capital Amman, where there is a heavy influence of Levantine Arabic and patterns of urban immigration, in addition, it is spoken mainly in the capital Amman. Secondly, the Fallahi (agricultural) dialect is connected to rural and farming societies. On the other hand, the Bedouin dialect, which is used by nomadic and semi-nomadic tribes, preserves archaic features of Classical Arabic. Lastly, the Karaki dialect, which originates from the southern city of Karak, and is characterized by phonological features that distinguish it from all other Jordanian dialects.

Automatic dialect identification (ADI) is the task of identifying the dialect of a given text or utterance. Good ADI is a necessary condition for building a robust dialect-aware NLP pipeline. In this paper we represent the design, implementation and evaluation of a text of a text-based ADI system for Jordanian Arabic. It utilizes character-level n-gram features and a Linear Support Vector Classifier. The system uses character-level n-gram features and a Linear Support Vector Classifier. The system demonstrates that light-weight machine learning approaches can do well on this demanding multi-class classification problem.

II. RELATED WORK

Arabic dialect identification has gained increasing attention in the NLP community. One of the first large-scale corpora of Arabic dialects was released by Zaidan and Callison-Burch [2], and was used for data-driven classification of Egyptian, Gulf, Levantine, and Iraqi Arabic. Token level dialect identification for Gypitian and MSA using morphological features was introduced by Elfardy and Diab [3]. [4] has shown that character n-gram language models provide state-of-the-art results for country-level dialect classification over 25, which further demonstrates the importance of Arabic dialects.

Specifically, the dialectal situation of Jordanian Arabic has been studied less than that of Egyptian or Gulf Arabic. Jerrar et al. [5] developed the Curras corpus, which is a morphologically annotated Palestinian Arabic corpus developed the Curras corpus, a morphologically annotated Palestinian Arabic corpus with common features associated with Jordanian dialects. Support

Vector Machines (SVMs) utilizing n-gram features have always been competitive across Arabic NLP benchmarks [6]. Character-level models are specifically appropriate for dialectal Arabic due to their ability to naturally model morphological variation, code-switching phenomena, as well as dialect-specific phonological transliterations without the need for dialect-specific tokenization or morphological analyzers.

III. DATASET

A. Data Acquisition and Structure

The dataset is supplied as an Excel file (FullData.xlsx) including 9,555 rows and three columns: text (the Arabic dialect sample), Dialect (the dialect label), and confidence (the annotator’s confidence level: high, medium, or low), dialect The text samples, which cover a wide spectrum of daily subjects such as hobbies, social interactions, expressions, and brief informal sentences and phrases usually of regular Jordanian Arabic communication.

B. Label Normalization

The dialect labels in the raw dataset had inconsistent capitalization (for example, “ammani” and “Ammani” were classified as separate categories, same scenario when it came to “falahi” and “Fallahi”). Four unified dialect categories: Ammani, Fallahi, Bedouin, as well as Karaki were generated using label normalization, which involved mapping all lowercase variations to their canonical title-case forms.

C. Deduplication

The dataset was decreased from 9,555 to 8,577 unique samples (978 duplicates deleted, or roughly 10.2% of the original data) after duplicate entries were extracted and then removed using pandas drop_duplicates(). The final dialect distribution following deduplication is displayed in Table I.

TABLE I. Dataset Distribution After Preprocessing

Dialect	Samples (after dedup.)	Proportion
Ammani	2,856	33.3%
Fallahi	2,699	31.5%
Karaki	1,982	23.1%
Bedouin	1,040	12.1%
Total	8,577	100%

The dataset exhibits a major class imbalance where 33.3% of samples are Ammani, whereas 12.1% only are Bedouin. This disparity supports the use of class-balanced training objectives, and also has direct effects on model training.

IV. METHODOLOGY

A. Preprocessing text

Prior to feature extraction, each sample of raw Arabic text was cleaned and normalized using a proprietary workflow. The pipeline is made up of the following sequentially applied steps:

- 1) **Diacritical removal:** Arabic vocalization marks (tashkeel), shadda, and superscript alef (Unicode range U+064B–U+065F, U+0610–U+061A, U+0670) were removed since they add noise and are applied inconsistently in informal language.
- 2) **Tatweel removal:** To eliminate stylistic lengthening, the Arabic elongation character (U+0640) was removed.
- 3) **Removal of non-Arabic characters:** This includes the elimination of punctuation, Latin numerals, and characters, alongside all characters outside the Arabic Unicode block (U+0600–U+06FF) and whitespace.
- 4) **Alef normalization:** To decrease orthographic variation, all Alef variants (ا, آ, إ, ؤ) were combined to the plain Alef (ا).
- 5) **Repeated character normalization:** Three or more instances of the same character were merged into two instances, since in some dialects certain letters would be repeated.
- 6) **Whitespace normalization:** Both leading and trailing whitespace were eliminated, also multiple consecutive spaces were condensed into a single space.

B. Feature Extraction

The following configuration was utilized in order to extract features using scikit-learn’s TfidfVectorizer:

- **Analyzer:** Character-level n-grams retrieved within word borders using the char_wb analyzer (padding applied). This option avoids cross-word boundary artifacts while capturing intra-word character patterns.
- **N-gram range:** (2, 5) — Bigrams up to 5-grams are recovered, encompassing both the lengthier morphological features as well as short functional patterns typical of each dialect.
- **Maximum features:** 50,000 — To manage memory and computation, the vocabulary is limited to the 50,000 most common n-gram kinds.

- **Sublinear TF scaling:** True — In order to decrease the dominance of highly frequent phrases, term frequency is log-scaled ($1 + \log(\text{tf})$)
- **Minimum document frequency:** min_df=2 — to minimize noise from hapax legomena, N-grams that appear in less than two samples are removed.

To ensure that there was no data leakage, the TF-IDF vectorizer was only fitted on the training set and then applied to the test set without refitting.

C. Classification Model

The classification model is a scikit-learn Linear Support Vector Classifier (Linear SVC), which utilizes a one-vs-rest multi-class strategy to maximize linear SVM goal. Since LinearSVC grows effectively to tens of thousands of features, it is a generally good fit for high-dimensional sparse text features spaces. In addition to that, class_weight="balanced", which automatically alters the penalty for each class inversely proportional to its frequency in the training set, was utilized to configure the classifier. By giving minority classes a higher misclassification cost, this covers up for the dataset's imbalance, particularly the underrepresentation of Bedouin samples. Hence, to guarantee convergence, the maximum number of iterations was set at 2,000.

D. Training and Evaluation Protocol

For this step, stratified sampling alongside random_state=42 were used, the preprocessed dataset was divided into training (80%) and test (20%) sets while still maintaining the class distribution in both sections. Moreover, there were 6,861 samples in the training set and 1,716 samples in the test set. Using five-fold stratified cross-validation grid search over $C \in \{0.01, 0.1, 1, 10\}$, hyperparameter optimization was performed on the regularization parameter C of LinearSVC, maximizing macro-averaged F1-score. After that, the optimal model was assessed on the held-out test set upon being retrained on the whole training set.

V. RESULTS AND DISCUSSION

A. Classification Performance

Alongside macro and weighted averages, Table II illustrates the best model's per-class precision, recall and F1-score on the test set. The accuracy is 84.4% overall.

TABLE II. Classification Report — Best LinearSVC Model

Dialect	Precision	Recall	F1-Score	Support
Ammani	0.91	0.92	0.92	571
Bedouin	0.78	0.78	0.78	208
Fallahi	0.79	0.81	0.80	540
Karaki	0.85	0.81	0.83	397
Accuracy (overall)	—	—	0.84	1716
Macro Avg	0.83	0.83	0.83	1716
Weighted Avg	0.84	0.84	0.84	1716

Due to its greater presence in the training data (2,856) compared to other dialects, the Ammani dialect performed optimally, with an F1-score of 0.92. F1-score of 0.80 for the Fallahi dialect, meanwhile the Karaki dialect received the F1-score of 0.83, lastly a 0.78 F1-score for the Bedouin dialect representing the lowest performance, which is consistent with its considerably smaller training set (1,040 samples). This suggests that a non-trivial percentage of Bedouin samples are misclassified as dialects with more training data.

B. Confusion Analysis

The confusion matrix found on the test partition is displayed in Table III.

TABLE III. Confusion Matrix (Rows: Actual, Columns: Predicted)

Actual \ Pred.	Ammani	Bedouin	Fallahi	Karaki
Ammani	525	5	25	16
Bedouin	15	162	22	9
Fallahi	27	16	437	60
Karaki	25	8	44	320

Off-diagonal values illustrate a pattern of misclassification, on the other hand diagonal values show occurrences that were correctly classified. Fallahi-Karaki (60 Fallahi samples predicted as Karaki and 44 Karaki samples predicted as Fallahi). And Fallahi-Bedouin (27 Fallahi predicted as Bedouin, meanwhile 22 Bedouin predicted as Fallahi) are the most frequent misclassification pairs. These misinterpretations demonstrate the true linguistic closeness between these

dialects: Bedouin and Fallahi share some archaic lexical items and phonological characteristics, whilst Fallahi and Karaki share rural southern Jordanian terminology. Short, context-poor utterances and dialectally perplexing statements account for a disproportionate fraction of classification errors, based on error analysis of 268 incorrectly classified samples.

C. Hyperparameter Optimization

The ideal value was found by utilizing grid search over the regularization parameter C with five-fold stratified cross-validation (scoring: macro F1). The trade-off between maximizing the margin while also minimizing training error is controlled by the regularization parameter C . Stronger regularization is enforced by smaller values of C , which assists high-dimensional sparse feature spaces by decreasing overfitting. The final model evaluation described above employed the chosen optimal C value

VI. DISCUSSION

The findings illustrate how well character n -gram characteristics capture dialectal diversity in Jordanian Arabic. Character n -grams, in contrast to word-level characteristics, naturally encode morphological affixes, phonetic replacements, and spelling changes without the need for tokenizer or morphological analyzer tailored to a specific dialect. For example, the Ammani use of 'مشن' (mish, negation) versus Bedouin 'ما' (ma) and Karaki-specific suffixes are reliably captured by overlapping character sequences within these words.

The performance gap between Ammani ($F1 = 0.92$) and Bedouin ($F1 = 0.78$) is primarily attributable to class imbalance rather than inherent linguistic difficulty. The balanced class weighting partially mitigated this effect, but 1,040 Bedouin training samples remain insufficient relative to the 2,856 Ammani samples. Augmenting the Bedouin subset would be expected to yield the largest performance gains.

Frequent Fallahi–Karaki confusions suggest that these two dialects share a significant character-level signature, which is linguistically unsurprising given their geographic proximity. Distinguishing them may require lexical or semantic features that go beyond character patterns, or the incorporation of explicit dialect-specific lexicons.

VII. CONCLUSION AND FUTURE WORK

This paper presented an automatic Jordanian Arabic dialect identification system based on character n -gram

TF-IDF features and a balanced LinearSVC classifier. The system was evaluated on a dataset of 8,577 deduplicated labeled samples and achieved an overall accuracy of 84.4% and a macro-averaged F1-score of 0.83 across four dialect classes. The approach is computationally lightweight, requires no dialect-specific linguistic resources, and is readily deployable as a preprocessing component for downstream Arabic NLP pipelines.

Future work should focus on the following directions: (1) collection of additional Bedouin dialect samples to address the class imbalance and improve recall for the most under-represented class; (2) exploration of deep learning models such as AraBERT or CAMEL-BERT, which have demonstrated state-of-the-art performance on Arabic NLP benchmarks and may better capture long-range contextual dependencies; (3) incorporation of confidence-weighted training, leveraging the annotator confidence scores present in the dataset to down-weight uncertain labels during training.

REFERENCES

- [1] A. Versteegh, *The Arabic Language*. Edinburgh: Edinburgh University Press, 2014.
- [2] O. Zaidan and C. Callison-Burch, "Arabic dialect identification," *Computational Linguistics*, vol. 40, no. 1, pp. 171–202, 2014.
- [3] H. Elfardy and M. Diab, "Sentence level dialect identification in Arabic," in *Proc. 51st Annual Meeting of the Association for Computational Linguistics (ACL)*, Sofia, Bulgaria, 2013, pp. 456–461.
- [4] M. Salameh, H. Bouamor, and N. Habash, "Fine-grained Arabic dialect identification," in *Proc. 27th International Conference on Computational Linguistics (COLING)*, Santa Fe, NM, USA, 2018, pp. 1332–1344.
- [5] M. Jarrar, N. Habash, D. Alrimawi, D. Akra, and N. Zalmout, "Curras: An annotated corpus for the Palestinian Arabic dialect," *Language Resources and Evaluation*, vol. 51, no. 3, pp. 745–787, 2017.
- [6] A. Ali, N. Dehak, P. Cardinal, S. Khurana, S. Yella, J. Glass, P. Bell, and S. Renals, "Automatic dialect detection in Arabic broadcast speech," in *Proc. Interspeech*, San Francisco, CA, USA, 2016, pp. 2934–2938.
- [7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.